

Oppgaver noe lignende til oppg.1a ved skoleeksamen: **(Oppg.1b er ofte noe utskrift ifm. bruk av STL)**

Hva blir utskriften fra følgende program (litt hjelp: det blir 5 linjer):

```
#include <iostream>
#include <string>
using namespace std;

class A {
protected:
    string nvnl;    int dager;
public:
    A(string t, int d)    { nvnl = t;  dager = d;    }
    virtual void display() { cout << nvnl << " -> " << dager; }
    bool mindre(A &a)    { return (nvnl < a.nvnl);    }
};

class B : public A {
private:
    string nv2;
public:
    B(string t, string u, int d) : A(t, d)    { nv2 = u;    }
    void display()    { A::display(); cout << " -> " << nv2;    }
    bool har(string t) { return (nv1.find(t) < nv1.length() &&
                                nv2.find(t) < nv2.length());    }
    char hent() { return (nv1[nv1.length() - 2*nv2.length()]);    }
};

int main() {
    A* aa;  B* bb[2];
    bb[0] = new B("Sogndal", "Rjukan", 3);
    bb[1] = new B("Brekkestrand", "Odda", 1);
    aa = bb[0];    aa->display(); cout << '\n';
    aa = bb[1];    (*aa).display(); cout << '\n';
    cout << ((*aa).mindre(*bb[0])) ? "ASBEST" : "DECRA" << '\n';
    cout << ((bb[0]->har("d")) ? "LEKT" : "SLOYFE") << '\n';
    cout << bb[1]->hent() << '\n';
    return 0;
}
```

Løsning:

```
Sogndal -> 3 -> Rjukan
Brekkestrand -> 1 -> Odda
ASBEST
SLOYFE
k
```

Hva blir utskriften fra følgende program (litt hjelp: det blir 5 linjer):

```
#include <iostream>
#include <string>
using namespace std;

class A {
protected:
    string nvnl;
    int dager;
public:
    A(string t, int d)        { nvnl = t;  dager = d;  }
    virtual void display()    { cout << dager << ": " << nvnl;  }
    void display2()           { cout << nvnl;  }
    bool harIkke(string t)    { return (!(nvnl.find(t) < nvnl.length())); }
    virtual char hent()       = 0;
};

class B : public A {
private:
    string nvnl2;
public:
    B(string t, string u, int d) : A(u, d) { nvnl2 = t; }
    void display()               { A::display(); cout << ", " << nvnl2; }
    void display2()              { cout << nvnl2; }
    char hent()                  { int n1 = nvnl.length();
                                return (nvnl[(n1+nvnl2.length()) % n1 + 2]); }
};

int main() {
    A* aa; B* bb[3], *bbb;
    bb[0] = new B("Edinburgh", "The Place", 4);
    bb[2] = new B("Athen", "Arethusa", 1);
    bb[1] = new B("Armenistis", "Valeta", 14);

    aa = bb[1];    aa->display();    cout << '\n';
    aa = bb[0];    (*aa).display();  cout << '\n';
    bbb = (B*) aa;  bbb->display2();  cout << '\n';
    cout << "Haaland + " << (((*aa).harIkke("Place"))? "Haug": "MaEl") << '\n';
    aa = bb[2];    cout << aa->hent() << '\n';
    return 0;
}
```

Løsning:

```
14: Valeta, Armenistis
4: The Place, Edinburgh
Edinburgh
Haaland + MaEl
a
```

Hva blir utskriften fra følgende program (litt hjelp: det blir 5 linjer):

```
#include <iostream>
#include <string>
using namespace std;

class A {
private:
    string txt1;
    int num;
public:
    A(string t, int n) { txt1 = t; num = n; }
    virtual void display() { cout << txt1 << ' ' << num << ' '; }
    virtual bool ulik(string t) { return (txt1 != t); }
};

class B : public A {
private:
    string txt2, txt3;
public:
    B(string t1, string t2, string t3, int num) : A(t3, num)
    { txt2 = t2; txt3 = t1; }
    virtual void display()
    { A::display(); cout << txt2 << ' ' << txt3 << ' '; }
    virtual bool ulik(string t)
    { return (txt2 != t && txt3 != t); }
    bool mindre(B & b)
    { return (txt2 < b.txt2 || txt3 < b.txt3); }
};

int main() {
    B* bobj1 = new B("Everton", "ManC", "Cardiff", 1);
    bobj1->display(); cout << '\n';
    A* bobj2 = new B("Norwich", "Ipswich", "Leeds", 2);
    bobj2->display(); cout << '\n';
    cout << (((*bobj1).ulik("ManC")) ? "Gdansk" : "Sopot") << '\n';
    cout << (((*bobj2).ulik("Leeds")) ? "Armenistis" : "Valeta") << '\n';
    cout << (((*bobj1).mindre(*(B*)bobj2)) ? "ME" : "IN") << '\n';
    return 0;
}
```

Løsning:

Cardiff 1 ManC Everton
Leeds 2 Ipswich Norwich
Sopot
Armenistis
ME